Specialized products formed from tyrosine

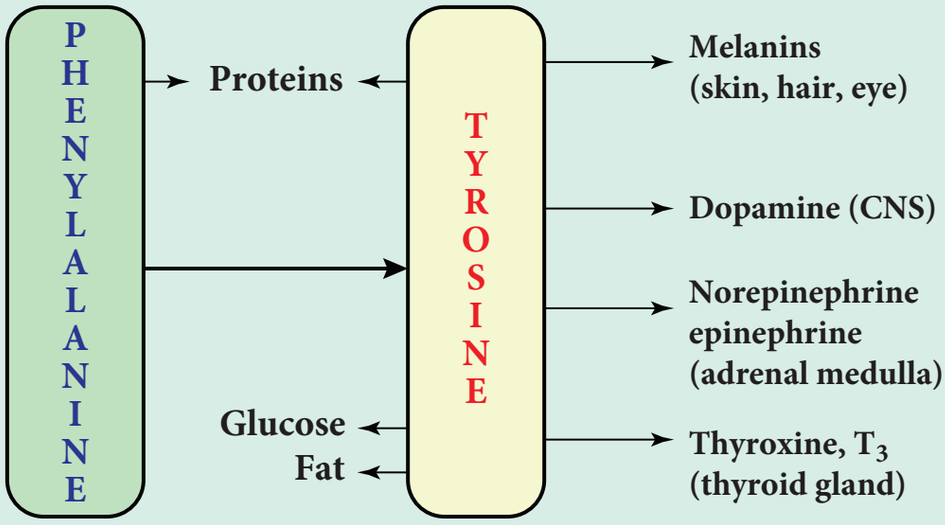

Figure 4.13 Specialized products formed form amino acids

Many specialized products are formed from tyrosine in different tissues (Figure 4.13). In the nervous system, tyrosine serves as the precursor for dopamine. In the adrenal glands, it is converted to stress hormones namely, epinephrine and nor-epinephrine. In the thyroid gland, tyrosine is iodinated and converted to thyroid hormones namely tri-iodo thyronine (T3) and tetraiodothyronine (T4-thyroxine)

4.4 Formation of Melanin

Melan means black. Melanin is the pigment that is responsible for the colour of the skin, hair and eyes. Melanin is a group of random polymers of indole quinone formed from tyrosine in the melanocytes. The major enzyme involved in its synthesis is Tyrosinase, which is a copper dependent enzyme (Figure 4.14).

1. Tyrosine is hydroxylated to form dihydroxyphenyl alanine (DOPA)

2. DOPA is decarboxylated to DOPA quinone.

3. DOPA quinone is converted to indole quinone by a series of reactions.

4. Indole quinone polymerizes to melanin.

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